A Appendix: Proofs

A.1 Proof of Theorem 4.1

Proof. Recall that

$$\widehat{\Gamma} = \frac{1}{n}\sum_{i=1}^{n} B(X_i)B(X_i)^\top, \qquad \Gamma = E\big[B(X)B(X)^\top\big],$$

and define the scaled matrices

$$\widehat{\Gamma}^\star = K_n\widehat{\Gamma}, \qquad \Gamma^\star = K_n\Gamma.$$

By the assumed Gram scaling condition (4.14), there exist constants $0 < c_0 < C_0 < \infty$, independent of n, such that

$$c_0 \le \lambda_{\min}(\Gamma^\star) \le \lambda_{\max}(\Gamma^\star) \le C_0.$$

Hence, the population scaled Gram matrix is uniformly positive definite and has uniformly bounded operator norm.

We first establish convergence of the empirical Gram matrix. For each pair of basis indices j, k, write

$$\widehat{\Gamma}_{jk} - \Gamma_{jk} = \frac{1}{n}\sum_{i=1}^{n}\{B_j(X_i)B_k(X_i) - E[B_j(X)B_k(X)]\}.$$

Since the B-spline basis satisfies the boundedness condition (4.15), the products $B_j(X_i)B_k(X_i)$ are uniformly bounded. Therefore, Bernstein's inequality yields, for every $t > 0$,

$$\Pr\left(\left|\widehat{\Gamma}_{jk} - \Gamma_{jk}\right| > t\right) \le 2\exp\{-C_1 n t^2\}$$

for sufficiently small t, where $C_1 > 0$ is independent of j, k and n.

There are K_n^2 entries in the Gram matrix. Applying the union bound gives

$$\Pr\left(\max_{1\le j,k\le K_n}\left|\widehat{\Gamma}_{jk} - \Gamma_{jk}\right| > t\right) \le 2K_n^2\exp\{-C_1 n t^2\}.$$

Taking

$$t = C_2\sqrt{\frac{K_n\log n}{n}}$$

with a sufficiently large constant C_2, and using the fact that the spline dimension satisfies $K_n\log n/n \to 0$, gives

$$\max_{j,k}\left|\widehat{\Gamma}_{jk} - \Gamma_{jk}\right| = O_{\mathbb{P}}\left(\sqrt{\frac{\log n}{n}}\right).$$

Because the B-spline basis is locally supported, each row of the Gram matrix contains only a uniformly bounded number of nonzero population contributions. The same locality property applies to the empirical Gram matrix. Consequently, the operator norm can be controlled by the corresponding maximal entrywise deviation together with the effective spline dimension, giving

$$\left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| = O_{\mathbb{P}}\left(\sqrt{\frac{K_n\log n}{n}}\right).$$

Thus,

$$\left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| \xrightarrow{\mathbb{P}} 0.$$

We next apply Weyl's eigenvalue perturbation inequality. For symmetric matrices M and N,

$$|\lambda_j(M) - \lambda_j(N)| \le \|M - N\|,$$

for every eigenvalue index j. Hence,

$$\left|\lambda_{\min}(\widehat{\Gamma}^\star) - \lambda_{\min}(\Gamma^\star)\right| \le \left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| = o_{\mathbb{P}}(1),$$

and similarly,

$$\left|\lambda_{\max}(\widehat{\Gamma}^\star) - \lambda_{\max}(\Gamma^\star)\right| \le \left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| = o_{\mathbb{P}}(1).$$

It follows that

$$\lambda_{\min}(\widehat{\Gamma}^\star) = \lambda_{\min}(\Gamma^\star) + o_{\mathbb{P}}(1)$$

and

$$\lambda_{\max}(\widehat{\Gamma}^\star) = \lambda_{\max}(\Gamma^\star) + o_{\mathbb{P}}(1).$$

Since the eigenvalues of $\Gamma^\star$ are uniformly bounded away from zero and infinity, these relations establish the claimed uniform eigenvalue convergence.

In particular, let $c = c_0/2$. On the event

$$\left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| < \frac{c_0}{2},$$

Weyl's inequality gives

$$\lambda_{\min}(\widehat{\Gamma}^\star) \ge \lambda_{\min}(\Gamma^\star) - \left\|\widehat{\Gamma}^\star - \Gamma^\star\right\| \ge \frac{c_0}{2} = c.$$

Because the probability of this event tends to one,

$$\Pr\left\{\lambda_{\min}(\widehat{\Gamma}^\star) \ge c\right\} \longrightarrow 1.$$

Therefore, with probability tending to one, $\widehat{\Gamma}^\star$ is positive definite and hence nonsingular. Since

$$\widehat{\Gamma} = K_n^{-1}\widehat{\Gamma}^\star,$$

and K_n is finite and strictly positive for every n, multiplication by K_n^{-1} does not change nonsingularity. Consequently,

$$\Pr\left\{\widehat{\Gamma} \text{ is nonsingular}\right\} \longrightarrow 1.$$

This completes the proof. □

A.2 Proof of Theorem 5.1

Proof. Let

$$V_i = Z_i - \mu_Z(X_i), \qquad \delta_{Yi} = \widehat{\mu}_Y(X_i) - \mu_Y(X_i), \qquad \delta_{Zi} = \widehat{\mu}_Z(X_i) - \mu_Z(X_i).$$

Since

$$Y_i = AZ_i + m(X_i) + \varepsilon_i$$

and

$$\mu_Y(X_i) = A\mu_Z(X_i) + m(X_i),$$

we have

$$Y_i - \mu_Y(X_i) = AV_i + \varepsilon_i.$$

Consequently, the profile residuals can be written as

$$\widetilde{Z}_i = Z_i - \widehat{\mu}_Z(X_i) = V_i - \delta_{Zi},$$

and

$$\widetilde{Y}_i = Y_i - \widehat{\mu}_Y(X_i) = AV_i + \varepsilon_i - \delta_{Yi}.$$

Equivalently,

$$\widetilde{Y}_i = A\widetilde{Z}_i + \varepsilon_i + R_i, \qquad R_i = A\delta_{Zi} - \delta_{Yi}.$$

By the definition of the profile estimator,

$$\widehat{A} = \left(\sum_{i=1}^n \widetilde{Y}_i\widetilde{Z}_i^\top\right)\left(\sum_{i=1}^n \widetilde{Z}_i\widetilde{Z}_i^\top\right)^{-1}.$$

Therefore,

$$\widehat{A} - A = \left\{\frac{1}{n}\sum_{i=1}^n \widetilde{Y}_i\widetilde{Z}_i^\top - A\frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i\widetilde{Z}_i^\top\right\}\left\{\frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i\widetilde{Z}_i^\top\right\}^{-1}.$$

Using

$$\widetilde{Y}_i - A\widetilde{Z}_i = \varepsilon_i + R_i,$$

we obtain

$$\widehat{A} - A = \left\{\frac{1}{n}\sum_{i=1}^n (\varepsilon_i + R_i)\widetilde{Z}_i^\top\right\}\widehat{\Omega}_n^{-1},$$

where

$$\widehat{\Omega}_n = \frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i\widetilde{Z}_i^\top.$$

Thus,

$$\sqrt{n}(\widehat{A} - A) = \left\{\frac{1}{\sqrt{n}}\sum_{i=1}^n \varepsilon_i\widetilde{Z}_i^\top + \frac{1}{\sqrt{n}}\sum_{i=1}^n R_i\widetilde{Z}_i^\top\right\}\widehat{\Omega}_n^{-1}. \tag{A.1}$$

We first consider the denominator. Since

$$\widetilde{Z}_i = V_i - \delta_{Zi},$$

we have

$$\widehat{\Omega}_n = \frac{1}{n}\sum_{i=1}^{n} V_i V_i^\top - \frac{1}{n}\sum_{i=1}^{n} V_i \delta_{Zi}^\top - \frac{1}{n}\sum_{i=1}^{n} \delta_{Zi} V_i^\top + \frac{1}{n}\sum_{i=1}^{n} \delta_{Zi}\delta_{Zi}^\top.$$

By the law of large numbers,

$$\frac{1}{n}\sum_{i=1}^{n} V_i V_i^\top \xrightarrow{\mathbb{P}} E(VV^\top) = \Omega.$$

The assumed $L^2(P_X)$ consistency of the nuisance estimator implies

$$\|\widehat{\mu}_Z - \mu_Z\|_{L^2(P_X)} = o_{\mathbb{P}}(1).$$

Under the empirical-process conditions imposed in (5.19), the remaining terms satisfy

$$\frac{1}{n}\sum_{i=1}^{n} V_i \delta_{Zi}^\top = o_{\mathbb{P}}(1), \qquad \frac{1}{n}\sum_{i=1}^{n} \delta_{Zi} V_i^\top = o_{\mathbb{P}}(1),$$

and

$$\frac{1}{n}\sum_{i=1}^{n} \delta_{Zi}\delta_{Zi}^\top = o_{\mathbb{P}}(1).$$

Hence

$$\widehat{\Omega}_n \xrightarrow{\mathbb{P}} \Omega. \tag{A.2}$$

Since Ω is positive definite,

$$\lambda_{\min}(\Omega) > 0.$$

It follows from (A.2) and continuity of matrix inversion on the set of nonsingular matrices that

$$\widehat{\Omega}_n^{-1} \xrightarrow{\mathbb{P}} \Omega^{-1}.$$

In particular, $\widehat{\Omega}_n$ is nonsingular with probability tending to one.

We next consider the numerator in (A.1). Since

$$\widetilde{Z}_i = V_i - \delta_{Zi},$$

we have

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varepsilon_i \widetilde{Z}_i^\top = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varepsilon_i V_i^\top - \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varepsilon_i \delta_{Zi}^\top.$$

The second term is a nuisance-estimation remainder. The stipulated empirical-process conditions in (5.19), together with

$$\|\widehat{\mu}_Z - \mu_Z\|_{L^2(P_X)} = o_{\mathbb{P}}(n^{-1/4}),$$

ensure that

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \varepsilon_i \delta_{Zi}^\top = o_{\mathbb{P}}(1). \tag{A.3}$$

Similarly, since

$$R_i = A\delta_{Zi} - \delta_{Yi},$$

the corresponding profile remainder satisfies

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} R_i \widetilde{Z}_i^\top = o_{\mathbb{P}}(1). \tag{A.4}$$

The latter is precisely the remainder condition imposed in (5.19). Notice that the argument only requires the nuisance estimators to converge sufficiently fast in $L^2(P_X)$; it does not require the incorrect uniform condition

$$\sup_{x \in \mathcal{X}} \|\widehat{\mu}(x) - \mu(x)\| = o_{\mathbb{P}}(n^{-1/2}).$$

Combining (A.1), (A.3), and (A.4), we obtain

$$\sqrt{n}(\widehat{A} - A) = \left\{ \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varepsilon_i V_i^\top \right\} \Omega^{-1} + o_{\mathbb{P}}(1), \tag{A.5}$$

where the convergence of $\widehat{\Omega}_n^{-1}$ to Ω^{-1} has been used.

To obtain the limiting distribution, vectorize (A.5). With the column-wise vectorization operator,

$$\operatorname{vec}(\varepsilon_i V_i^\top) = (V_i \otimes I_q)\varepsilon_i.$$

Therefore,

$$\sqrt{n}\, \operatorname{vec}(\widehat{A} - A) = (\Omega^{-1} \otimes I_q) \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \operatorname{vec}(\varepsilon_i V_i^\top) + o_{\mathbb{P}}(1).$$

The summands have mean zero because

$$E(\varepsilon_i \mid X_i, Z_i) = 0,$$

and, by the assumed finite second moment,

$$E \left\| \varepsilon_i V_i^\top \right\|^2 < \infty.$$

Their covariance matrix is

$$\begin{aligned} \Xi &= \operatorname{Var}\big\{ \operatorname{vec}(\varepsilon V^\top) \big\} \\ &= E\left[(V \otimes I_q) \Sigma(X) (V^\top \otimes I_q) \right] \\ &= E\left[V V^\top \otimes \Sigma(X) \right]. \end{aligned}$$

Since the observations are independent and identically distributed, the multivariate central limit theorem gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \operatorname{vec}(\varepsilon_i V_i^\top) \xrightarrow{d} N(0, \Xi).$$

Finally, Slutsky's theorem yields

$$\sqrt{n}\, \operatorname{vec}(\widehat{A} - A) \xrightarrow{d} N(0, \Sigma_A),$$

where

$$\Sigma_A = (\Omega^{-1} \otimes I_q) \Xi (\Omega^{-1} \otimes I_q).$$

Since

$$V = Z - \mu_Z(X),$$

the covariance matrix Ξ can equivalently be written as

$$\Xi = E\left[\{Z - \mu_Z(X)\} \{Z - \mu_Z(X)\}^\top \otimes \Sigma(X) \right].$$

This proves (5.26)–(5.28). □

A.3 Proof of Theorem 6.1

Proof. Let

$$\widetilde{Z}_i^0 = Z_i - \mu_Z(X_i)$$

denote the population residualized covariate and define

$$\delta_{Zi} = \widehat{\mu}_Z(X_i) - \mu_Z(X_i), \qquad \delta_{Yi} = \widehat{\mu}_Y(X_i) - \mu_Y(X_i).$$

Then

$$\widetilde{Z}_i = Z_i - \widehat{\mu}_Z(X_i) = \widetilde{Z}_i^0 - \delta_{Zi}.$$

Furthermore, since

$$\mu_Y(X_i) = A\mu_Z(X_i) + m(X_i),$$

the regression model implies

$$Y_i - \mu_Y(X_i) = A\widetilde{Z}_i^0 + \varepsilon_i.$$

Consequently,

$$\widetilde{Y}_i = Y_i - \widehat{\mu}_Y(X_i) = A\widetilde{Z}_i^0 + \varepsilon_i - \delta_{Yi}.$$

Writing

$$R_i = A\delta_{Zi} - \delta_{Yi},$$

we may equivalently express the profile equation as

$$\widetilde{Y}_i = A\widetilde{Z}_i + \varepsilon_i + R_i.$$

By the definition of the profile least-squares estimator,

$$\widehat{A} = \left(\frac{1}{n}\sum_{i=1}^n \widetilde{Y}_i\widetilde{Z}_i^\top\right)\left(\frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i\widetilde{Z}_i^\top\right)^{-1}.$$

Hence

$$\widehat{A} - A = \left\{\frac{1}{n}\sum_{i=1}^n(\varepsilon_i + R_i)\widetilde{Z}_i^\top\right\}\widehat{\Omega}_n^{-1},$$

where

$$\widehat{\Omega}_n = \frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i\widetilde{Z}_i^\top.$$

Multiplication by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{A} - A) = \left\{\frac{1}{\sqrt{n}}\sum_{i=1}^n \varepsilon_i\widetilde{Z}_i^\top + \frac{1}{\sqrt{n}}\sum_{i=1}^n R_i\widetilde{Z}_i^\top\right\}\widehat{\Omega}_n^{-1}. \tag{A.6}$$

We first establish the convergence of the denominator. Since

$$\widetilde{Z}_i = \widetilde{Z}_i^0 - \delta_{Zi},$$

we have

$$\widehat{\Omega}_n = \frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i^0(\widetilde{Z}_i^0)^\top - \frac{1}{n}\sum_{i=1}^n \widetilde{Z}_i^0\delta_{Zi}^\top - \frac{1}{n}\sum_{i=1}^n \delta_{Zi}(\widetilde{Z}_i^0)^\top + \frac{1}{n}\sum_{i=1}^n \delta_{Zi}\delta_{Zi}^\top.$$

By the moment assumption and the law of large numbers,

$$\frac{1}{n}\sum_{i=1}^{n}\widetilde{Z}_i^0(\widetilde{Z}_i^0)^\top \xrightarrow{\mathbb{P}} E\left[\widetilde{Z}^0(\widetilde{Z}^0)^\top\right] = \Omega.$$

The nuisance-estimation conditions in Assumptions 6.1–6.6 imply

$$\frac{1}{n}\sum_{i=1}^{n}\widetilde{Z}_i^0\delta_{Zi}^\top = o_{\mathbb{P}}(1), \qquad \frac{1}{n}\sum_{i=1}^{n}\delta_{Zi}(\widetilde{Z}_i^0)^\top = o_{\mathbb{P}}(1),$$

and

$$\frac{1}{n}\sum_{i=1}^{n}\delta_{Zi}\delta_{Zi}^\top = o_{\mathbb{P}}(1).$$

Therefore,

$$\widehat{\Omega}_n \xrightarrow{\mathbb{P}} \Omega. \tag{A.7}$$

Since Ω is positive definite, $\widehat{\Omega}_n$ is nonsingular with probability tending to one and

$$\widehat{\Omega}_n^{-1} \xrightarrow{\mathbb{P}} \Omega^{-1}.$$

Next, decompose the leading stochastic term as

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i\widetilde{Z}_i^\top = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i(\widetilde{Z}_i^0)^\top - \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i\delta_{Zi}^\top.$$

By the empirical-process and nuisance-rate conditions in Assumption 6.6,

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i\delta_{Zi}^\top = o_{\mathbb{P}}(1).$$

Similarly, the profile remainder condition (5.19) gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}R_i\widetilde{Z}_i^\top = o_{\mathbb{P}}(1).$$

Substituting these relations into (A.6) and using (A.7), we obtain

$$\sqrt{n}(\widehat{A} - A) = \left\{\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\varepsilon_i(\widetilde{Z}_i^0)^\top\right\}\Omega^{-1} + o_{\mathbb{P}}(1). \tag{A.8}$$

Applying the column-wise vectorization operator and using

$$\operatorname{vec}(\varepsilon_i(\widetilde{Z}_i^0)^\top) = (\widetilde{Z}_i^0 \otimes I_q)\varepsilon_i,$$

we obtain

$$\sqrt{n}\,\operatorname{vec}(\widehat{A} - A) = (\Omega^{-1} \otimes I_q)\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\operatorname{vec}\left[\varepsilon_i(\widetilde{Z}_i^0)^\top\right] + o_{\mathbb{P}}(1).$$

This proves the stochastic expansion (6.10).

It remains to derive the limiting distribution. Since

$$E(\varepsilon \mid X, Z) = 0,$$

we have

$$E\left[\operatorname{vec}\left\{\varepsilon(\widetilde{Z}^0)^\top\right\}\right] = 0.$$

Moreover, the moment condition in Assumption 6.1 ensures that the covariance matrix of the vectorized summand is finite. In particular,

$$\begin{aligned}\Xi &= \operatorname{Var}\left[\operatorname{vec}\left\{\varepsilon(\widetilde{Z}^0)^\top\right\}\right] \\ &= E\left[\left\{\widetilde{Z}^0(\widetilde{Z}^0)^\top\right\} \otimes \Sigma(X)\right].\end{aligned}$$

Thus, by the multivariate central limit theorem,

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \operatorname{vec}\left[\varepsilon_i(\widetilde{Z}_i^0)^\top\right] \xrightarrow{d} N(0, \Xi).$$

Combining this convergence with the expansion (6.10) and applying Slutsky's theorem yields

$$\sqrt{n}\,\operatorname{vec}(\widehat{A} - A) \xrightarrow{d} N(0, \Sigma_A),$$

where

$$\Sigma_A = (\Omega^{-1} \otimes I_q)\Xi(\Omega^{-1} \otimes I_q).$$

This establishes (6.11) and (6.12), and completes the proof. □